

EBN 111

Practical 1: Practical Assignment

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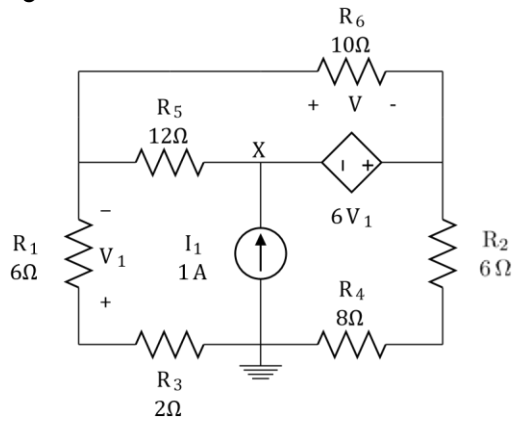
Name and Surname: _____

Student Number: _____

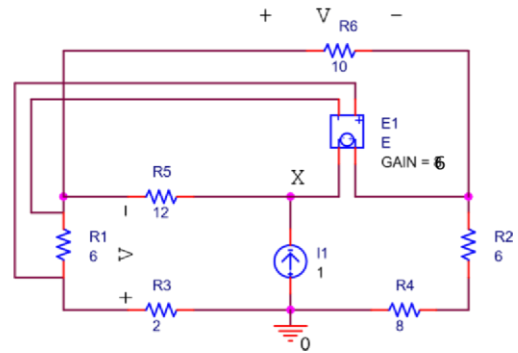
Instructions (PLEASE READ!!).

1. You need to complete this assignment during the practical session.
 2. After completing this assignment, log on to clickUP and fill in your answers online (Practical 1 Online Evaluation).
 3. Complete this assignment in groups of two. Communication between groups is strictly prohibited and can result in you being removed from (and subsequently failing) the practical.
- This form will not be collected, only the answers you submit to clickUP will be evaluated.
 - Give all answers with no metric prefix. For example, if you measure 523 mA, you need to convert the answer to its form with no prefix (0.523A).
 - Do not include units (V, A, etc.) as clickUP cannot evaluate these either. With the example above you would only type 0.52 into clickUP.
 - Use a full stop to denote a decimal, do not use a comma, as follows:
2.1 not 2,1
This is because clickUP cannot evaluate numbers using a comma.
 - Provide all answers to two decimal places.
 - Note that the values of Question 3.1.5 and Question 3.2.1 is selected from a random pool and may not be the same as that of your group member.

Question 1



(a) Circuit diagram



(b) Simulated circuit diagram

Figure 1: Circuit diagrams for Question 1

Capture the above circuit schematic (Figure 1) in OrCAD, and answer the questions that follow.

Hint

If you struggle to find the DC current source, the part name is `Idc`. Search for it under parts



Hint

If you struggle to find the voltage controlled voltage source, search for it by looking for the component named `E`. Pay particular attention to the polarity of this element!



Hint

Remember that the values underneath the resistors represent their resistances.

Questions

- 1.1) What is the voltage across R1? V

- 1.2) What is the voltage across R6? V

- 1.3) What is the current flowing through R2 and R4? A

- 1.4) What is the node voltage at node X? V

Question 2

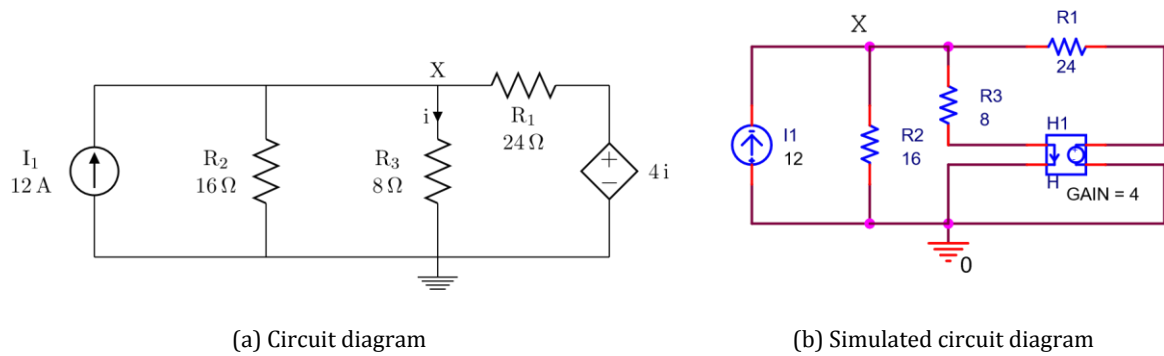
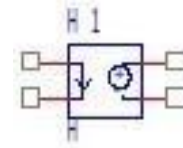


Figure 2: Circuit diagrams for Question 2

Capture the above circuit schematic (Figure 2) in OrCAD, and answer the questions that follow.

Hint

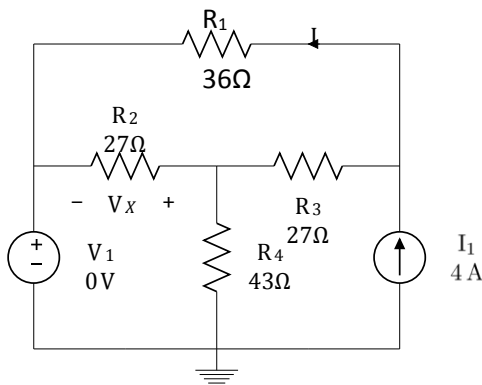
If you struggle to find the current controlled voltage source, search for it by looking for the component named H. Pay particular attention to the polarity of this component!



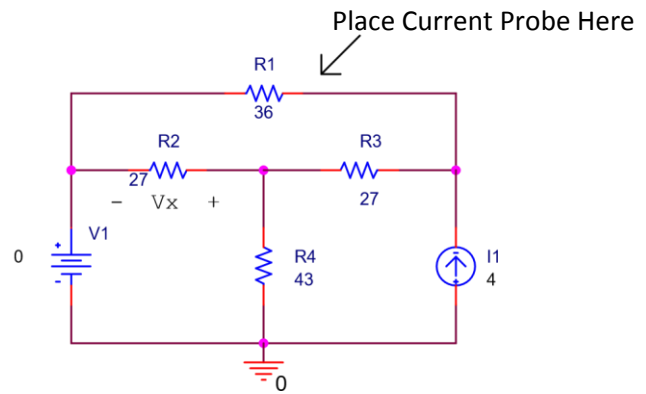
Questions

- 2.1) What is the node voltage at node X? _____ V
- 2.2) What is the current flowing through R1 resistor? _____ A
- 2.3) What is the total power absorbed? _____ W
- 2.4) What is the total power provided? _____ W

Question 3



(a) Circuit diagram



(b) Simulated circuit diagram

Figure 3: Circuit diagrams for Question 3

Capture the above circuit schematic (Figure 3) in OrCAD, and answer the questions that follow. When placing the current probe, please place it on the location indicated in Figure 3b (that is, to the right of R_1).

Questions

3.1) Using the "DC Sweep" analysis type sweep V_1 over the range 0 to 100 V in 5 V steps.

Hint

To display any signal after the DC sweep, select a probe and place it in the circuit where you would like the value to be read.



3.1.1) What is the **minimum** current, I , that flows through R_1 ? _____ A

3.1.2) What is the **maximum** current, I , that flows through R_1 ? _____ A

Hint

To take accurate measurements from the graph, use the Cursor tool. In the PSpice window you should see these icons.

Due to a bug in OrCAD 16.6 you will also need to use the blue icons to the right of the cursor tool to read the last value in the graph. If its a minimum, click on the 4th blue icon from the right, if its a maximum, click on the 3rd icon from the right.



- 3.1.3) What is the **minimum** value of the voltage V_x ? _____ V
- 3.1.4) What is the **maximum** value of the voltage V_x ? _____ V

The values in the following questions are randomly chosen on clickUP. More specifically, the points in the graph where readings must be made is different for each person, but the circuit is the same for every person.

- 3.1.5) What is the voltage value V_x if:

Question 3.1.5.a) to 3.1.5.e) are available on clickUp in the Practical 1 Online Evaluation (the clickUP test where you have to fill in your answers for this whole practical). It is marked as Questions 13 to 17.

Hints

Remember to use the correct voltage probes in order to measure voltage across a resistor.
Also note the polarity of V_x (refer to Figure 3).
Read the values of V_x off of the graph using the cursor tool.

- 3.2) For the circuit created in question 3.1 Figure 3, set keep $V_1 = 0V$. Perform another DC sweep, but this time use the current source I_1 as the sweep variable, and sweep it from 0 to 5 A in 0.2 A increments. Use a current probe to measure the current through R_1 .

- 3.2.1) What is the current, I , that flows through R_1 if:

Question 3.2.1.a) to 3.2.1.e) are available on clickUp in the Practical 1 Online Evaluation (the clickUP test where you have to fill in your answers for this whole practical). It is marked as Questions 18 to 22.

clickUP instructions

- Now log on to clickUP and fill in your answers in: **Practical 1 Online Evaluation**
- Remember to use the decimal point and not the comma for number separation and do not supply units!
- Provide all answers up to 2 decimal places.